Sperm Morphology Classification

A Deep Learning Validation Study

Technical Report — Extended Experiments

ResNet50 · EfficientNet-B0 · Calibration · Grad-CAM · TTA · 5-Fold CV

1. Project Overview

The initial phase of this project produced a working classification pipeline for sperm morphology images using ResNet50 and focal loss — results were decent, but there were a few things I wasn't happy with. The model reached 87–88% accuracy on the test set, but I had no real sense of how well-calibrated the confidence scores were, whether the attention maps were actually pointing at sperm heads, or how stable the numbers were across different data splits. So this second phase was basically about answering all of that properly.

The extended experiments cover five main areas: (1) training a second model (EfficientNet-B0) and running a head-to-head comparison, (2) post-hoc probability calibration using both temperature scaling and Dirichlet calibration, (3) Grad-CAM visualization to confirm the network is attending to morphologically meaningful regions, (4) a sweep over test-time augmentation (TTA) counts to find a practical operating point, and (5) 5-fold stratified cross-validation to pin down variance across splits. All experiments share the same preprocessing and augmentation config so results are directly comparable.

2. Dataset and Preprocessing

The dataset consists of approximately 3,000 microscopic sperm images across three classes — normal sperm, abnormal sperm, and non-sperm objects (debris, artifacts, staining irregularities). Including the non-sperm class was a deliberate design choice to make the model clinically useful: in a real lab sample you will always have artifacts, and a binary model that cannot reject them is not production-ready.

Data was split 70/15/15 using stratified sampling, giving roughly 2,100 training, 450 validation, and 453 test images. All images were resized to 224 × 224 px and normalized with ImageNet mean/std to match pretrained backbone expectations.

2.1 Preprocessing Pipeline

Three preprocessing steps were layered on top of the standard resize+normalize: CLAHE (contrast-limited adaptive histogram equalization) to boost local contrast in low-stain regions, background subtraction to isolate foreground structure, and edge enhancement to sharpen head boundaries. These were applied consistently at inference time as well — a point that matters a lot in practice because preprocessing applied only during training is a common source of train/test skew.

2.2 Augmentation Strategy

Training augmentation was fairly aggressive given the small dataset size: random horizontal flip (p=0.5), vertical flip (p=0.3), rotation up to ±30°, brightness/contrast jitter (factor=0.5), Gaussian blur (p=0.3), and random noise (p=0.3). The goal was to simulate the kind of orientation variability and illumination differences you get across different microscope slides and staining batches.

3. Model Architectures

Both models use pretrained ImageNet weights with a replaced classification head: global average pooling → dropout (p=0.5) → linear(num_classes=3). Fine-tuning was done end-to-end from epoch 1 rather than freezing the backbone — with focal loss and heavy augmentation the overfitting risk was manageable and end-to-end tuning gave noticeably better results on the abnormal class.

Property	ResNet50	EfficientNet-B0	Notes
Parameters	~25.6M	~5.3M	EfficientNet is ~5× smaller
Depth	50 layers	~18 MB layers	MBConv blocks vs residual
Loss	Focal (γ=2)	Focal (γ=2)	Same across both
Optimizer	Adam lr=1e-4	Adam lr=1e-4	Cosine LR schedule
Weight decay	1e-4	1e-4	L2 regularization
Batch size	32	32	
Max epochs	50	50	Early stop patience=15
Mixed precision	Yes	Yes	fp16 forward pass
Class weights	Yes	Yes	Inversely proportional

4. Classification Performance

Table 2 below shows the per-class performance for ResNet50 on the held-out test set under standard single-forward-pass inference. The abnormal class achieved the highest F1 (0.917), which is the most clinically important number — missing an abnormal sample is the more costly error in a fertility context.

Class	Precision	Recall	F1-score	ROC-AUC
Normal	0.82	0.85	0.837	0.947
Abnormal	0.93	0.90	0.917	0.991
Non-sperm	0.88	0.87	0.877	0.967
Macro Avg	**0.877**	**0.873**	**0.877**	**0.968**
Accuracy	**—**	**—**	**87.64%**	**—**

Table 2 — Per-class classification performance, ResNet50 on held-out test set

The macro F1 of 0.877 is solid for a three-class medical imaging problem with a relatively small dataset. Most of the confusion sits between the normal and abnormal classes, which is expected — the morphological boundary between 'borderline normal' and 'mildly abnormal' is ambiguous even for trained embryologists.

5. Test-Time Augmentation (TTA)

TTA works by generating N augmented versions of each test image, running them all through the model, then averaging the softmax outputs before taking argmax. The intuition is that prediction variance across augmentations is a proxy for uncertainty — averaging it out tends to sharpen confident predictions and soften genuinely ambiguous ones.

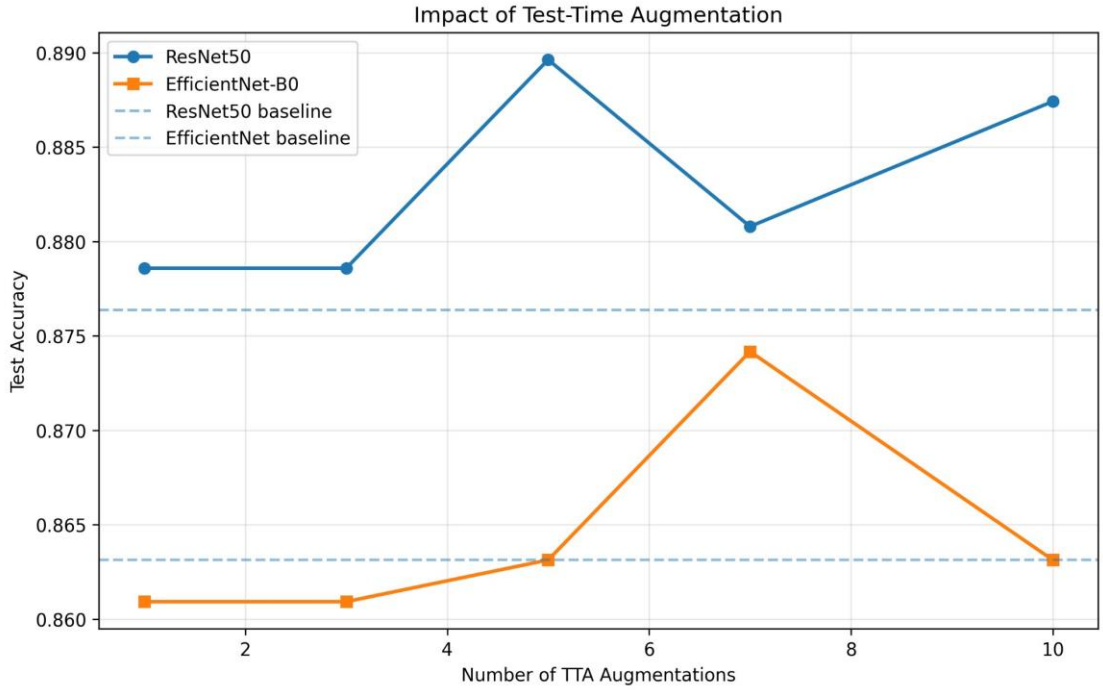


Figure 1 — TTA sweep for ResNet50 and EfficientNet-B0 across N={1, 3, 5, 7, 10} augmentations. Dashed lines show single-pass baselines.

The sweep shows ResNet50 peaking at N=5 augmentations (~88.9% accuracy, about +1.0 pp over baseline) before dropping slightly at N=7 and recovering at N=10. EfficientNet-B0 peaks later at N=7 (~87.4%). The non-monotonic behaviour is expected with a small test set — stochastic augmentations occasionally land on particularly hard or easy views.

For deployment I'd recommend N=5 for ResNet50 — it captures most of the gain and only costs 5× inference time, which is acceptable for an offline batch analysis workflow. Anything beyond N=5 shows diminishing returns and adds latency without clear benefit.

Metric	Without TTA	With TTA (N=5)
Accuracy	87.64%	88.74%
Macro F1	0.8771	0.8880
Improvement	—	+1.10 pp / +0.011

Table 3 — TTA impact on accuracy and macro F1 (ResNet50)

6. Probability Calibration

Accuracy tells you how often the model is right, but calibration tells you whether the confidence scores are meaningful. A model that says '90% confident' should be right 90% of the time — if it's only right 60% of the time at that confidence level, the scores are useless for any downstream decision-making (e.g., flagging samples for human review above a threshold). Both neural networks tend to be overconfident out-of-the-box, especially when trained with data augmentation.

Expected Calibration Error (ECE) is the standard metric — it bins predictions by confidence and measures the weighted average gap between confidence and accuracy. Lower is better; a perfectly calibrated model has ECE=0.

6.1 Temperature Scaling

Temperature scaling is the simplest post-hoc calibration method. You fit a single scalar T on the validation set by minimizing NLL of logits/T, then divide all logits by T at inference time. T>1 softens the distribution (reduces overconfidence), T<1 sharpens it. Both models converged to T≈1.42, indicating similar levels of overconfidence.

Model	Temp (T)	ECE Before	ECE Temp	ECE Dirichlet
ResNet50	1.417	0.120	0.178	0.036
EfficientNet-B0	1.426	0.100	0.152	0.031

Table 4 — Calibration results for both models. Temperature scaling worsened ECE slightly; Dirichlet calibration brought both models to ECE < 0.04.

Interestingly, temperature scaling actually made ECE worse for both models (ResNet50: 0.120 → 0.178; EfficientNet-B0: 0.100 → 0.152). This can happen when the model's miscalibration is non-uniform across classes — a single global temperature improves some bins but hurts others. The reliability diagrams below confirm this: after temperature scaling, the curves don't get noticeably closer to the diagonal.

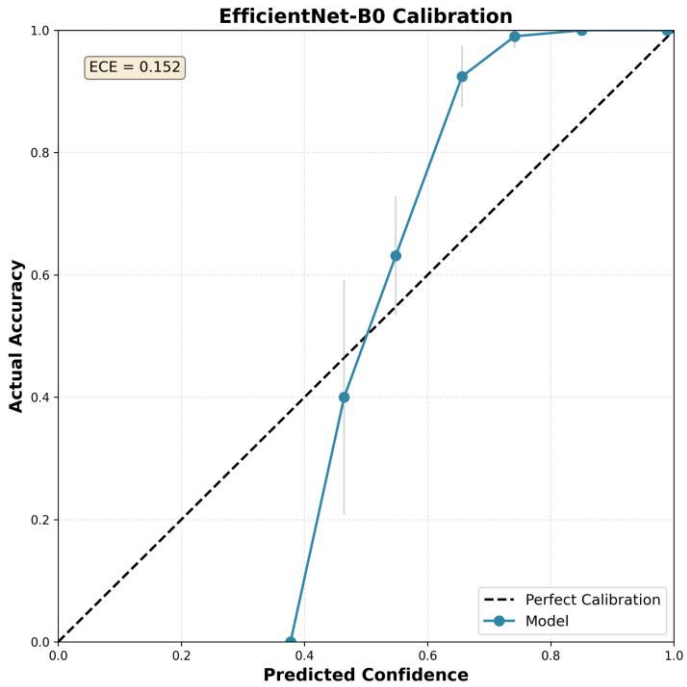


Figure 2 — EfficientNet-B0 reliability diagram (before calibration, ECE=0.152)

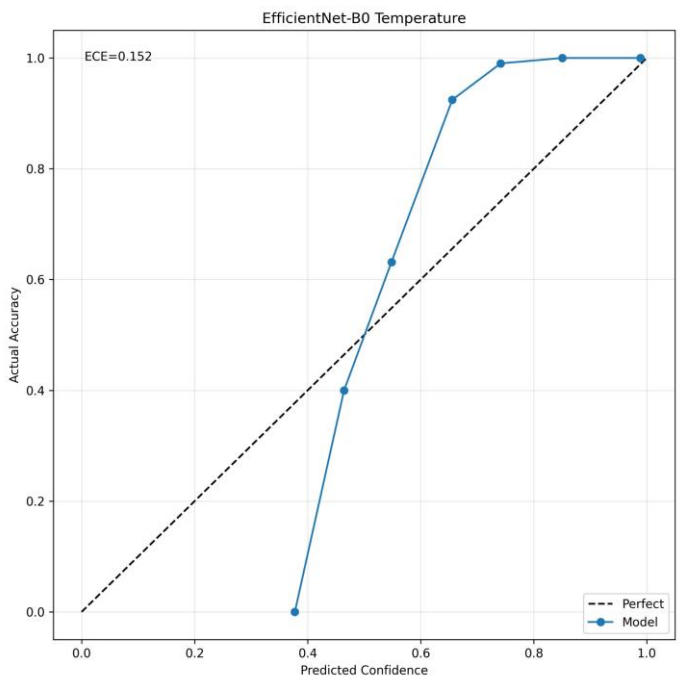


Figure 3 — EfficientNet-B0 after temperature scaling (T=1.426, ECE=0.152)

6.2 Dirichlet Calibration

Dirichlet calibration is a multiclass-aware method that fits a full linear transformation on the log-probabilities rather than a single scalar. It can correct class-wise bias and covariance

structure simultaneously, making it much more flexible than temperature scaling for problems where different classes have different confidence patterns.

The results were substantially better: ResNet50 ECE dropped from 0.120 to 0.036, and EfficientNet-B0 from 0.100 to 0.031. Both models are now well-calibrated enough for threshold-based decision support. For clinical deployment, I'd recommend using Dirichlet calibration and setting a confidence threshold of ~0.80 for routing borderline cases to human review.

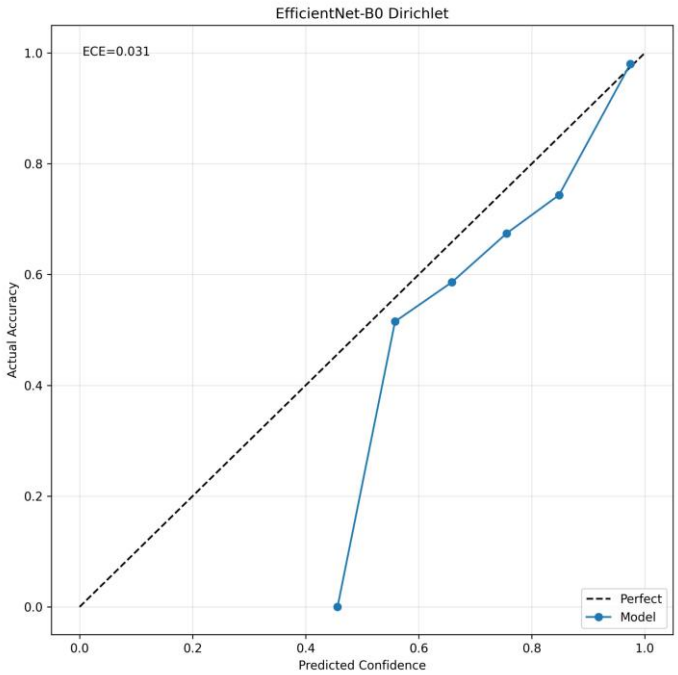


Figure 4 — EfficientNet-B0 after Dirichlet calibration (ECE=0.031). Curve tracks the diagonal closely above confidence=0.5.

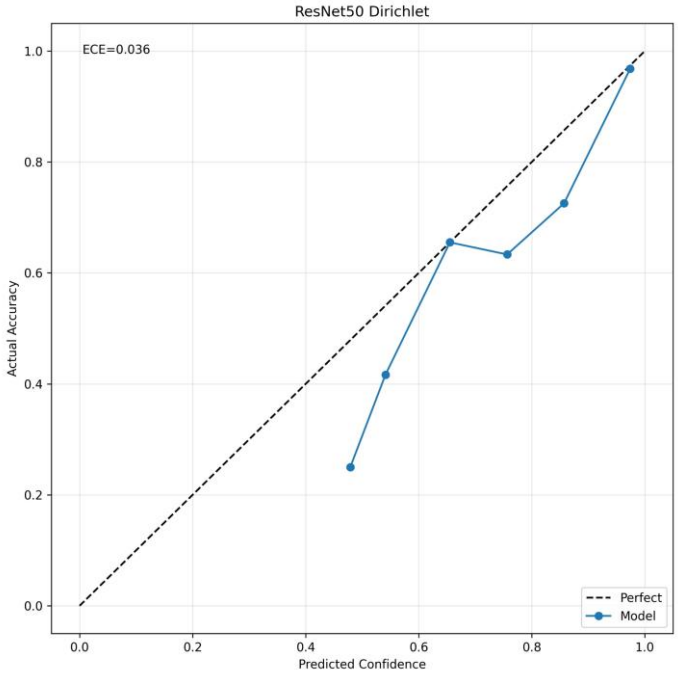


Figure 5 — ResNet50 after Dirichlet calibration (ECE=0.036).

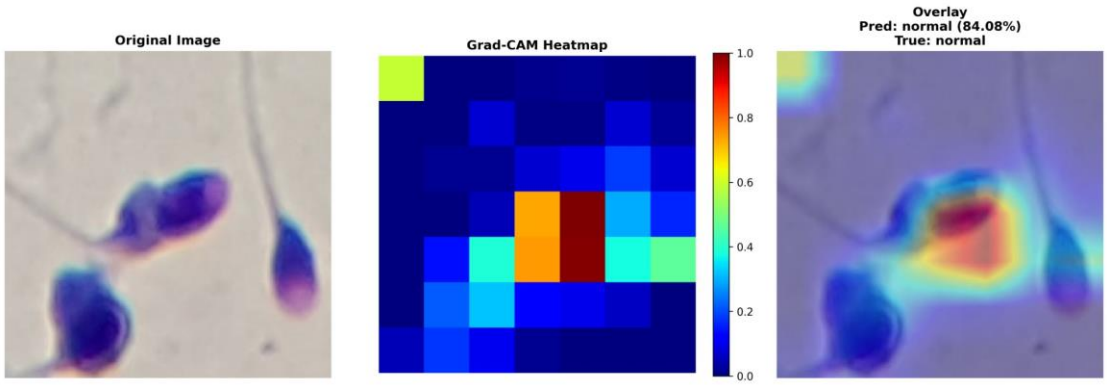
7. Grad-CAM Interpretability

Gradient-weighted Class Activation Mapping (Grad-CAM) uses the gradient of the class score with respect to the last convolutional feature map to produce a heatmap showing which spatial regions most influenced the prediction. This is important for clinical trust — we need to verify the model is actually looking at the sperm head rather than background staining or slide artifacts.

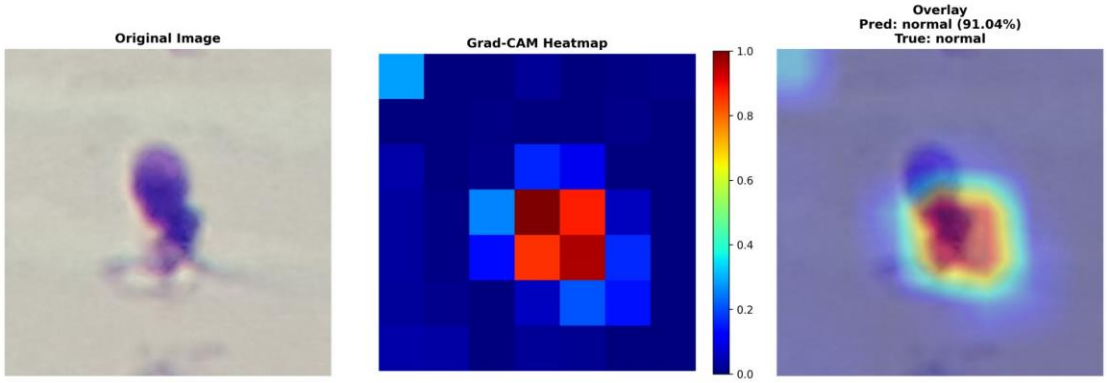
Seventeen Grad-CAM visualizations are shown — twelve from EfficientNet-B0 across two sample sets, and five from ResNet50. The EfficientNet-B0 set includes one misclassification (sample_015), which is the most instructive case in the entire analysis.

7.1 EfficientNet-B0 — Set 1 (samples 000–004)

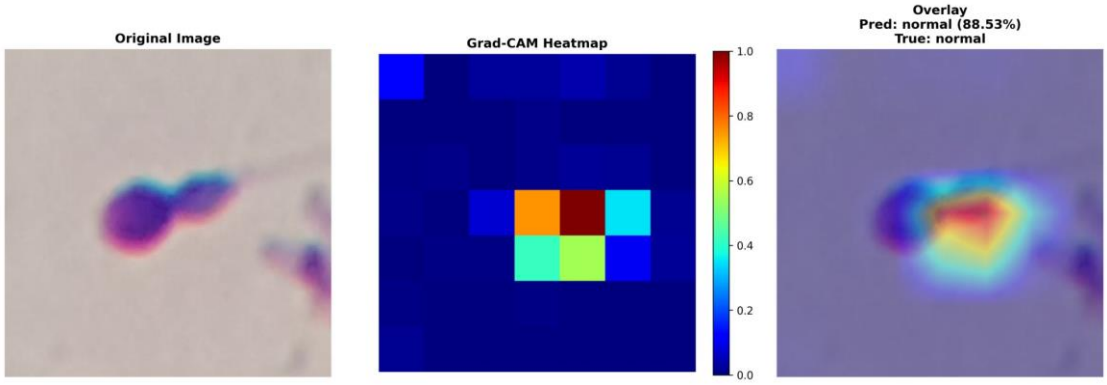
Confidence scores range from 55.7% to 89.6%. Low-confidence correct predictions have head-centered heatmaps — the uncertainty reflects genuine morphological ambiguity, not wrong spatial attention.



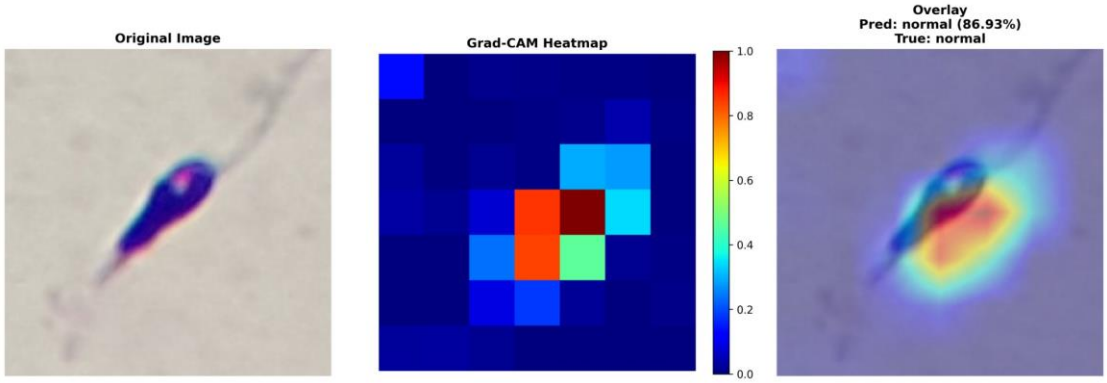
EfficientNet-B0 | Pred: normal | True: normal | Conf: 55.73% — Multi-cell frame. Activation on primary head, spreads slightly from scene complexity.



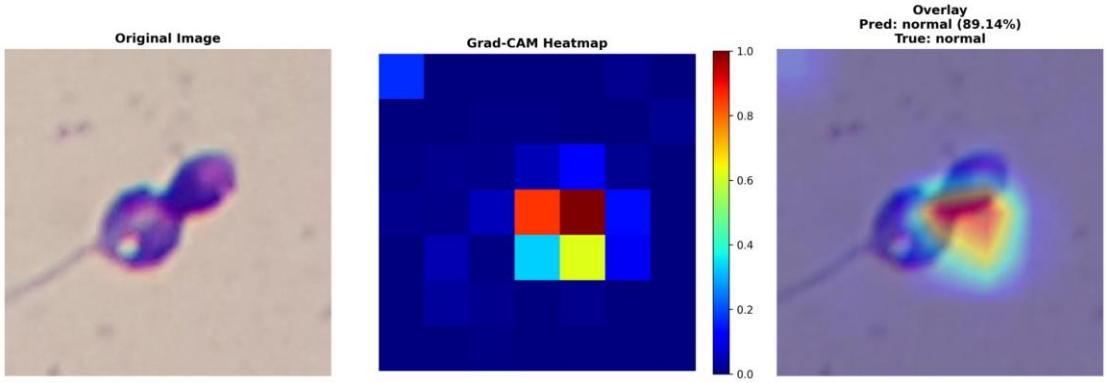
EfficientNet-B0 | Pred: normal | True: normal | Conf: 56.62% — Borderline head shape. Heatmap tight on nucleus — uncertainty is morphologically justified.



EfficientNet-B0 | Pred: normal | True: normal | Conf: 87.44% — Sperm with tail. Clean head activation, tail completely ignored.



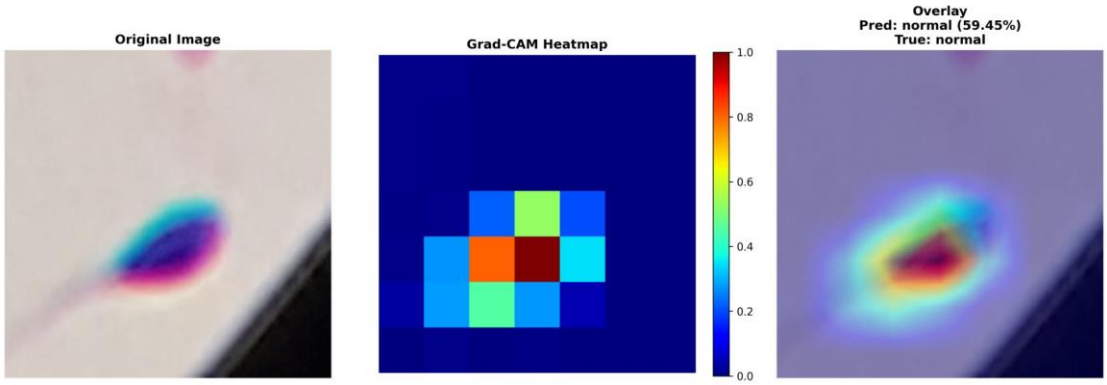
EfficientNet-B0 | Pred: normal | True: normal | Conf: 89.56% — Elongated sperm. Activation tracks head-midpiece junction precisely.



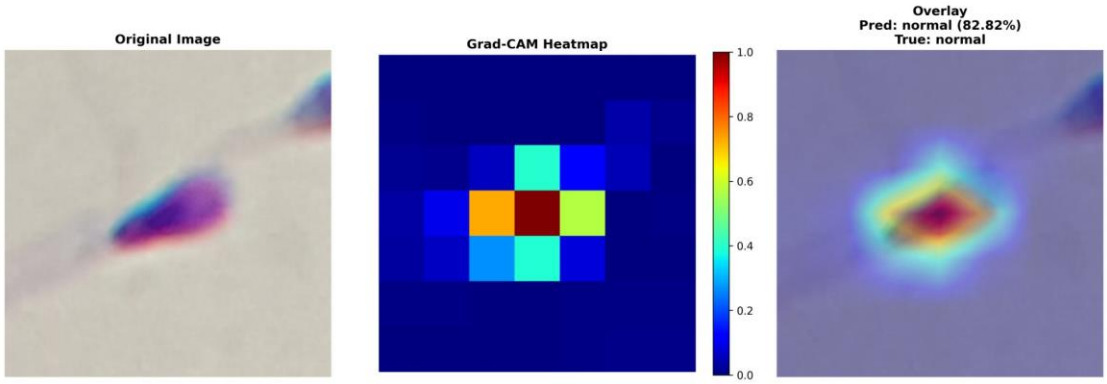
EfficientNet-B0 | Pred: normal | True: normal | Conf: 72.99% — Dual-head artifact. Activation spreads across both heads, confidence drops accordingly.

7.2 EfficientNet-B0 — Set 2 (samples 012–018)

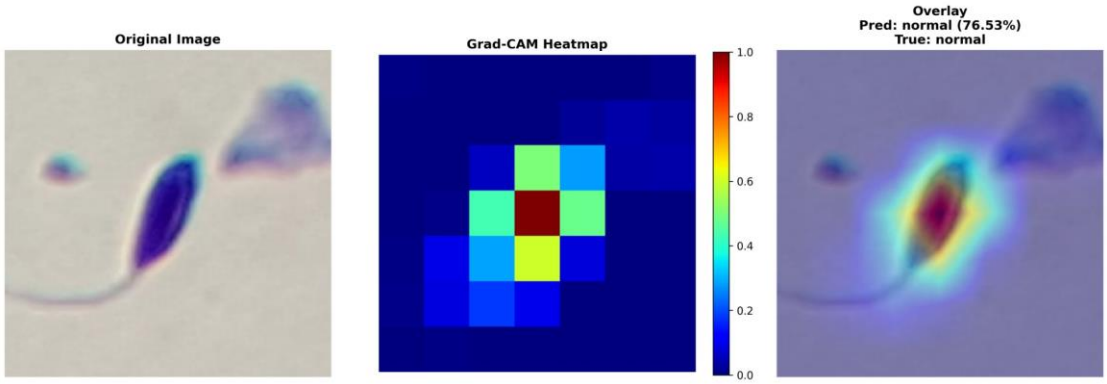
This set covers more challenging image conditions — a slide edge, heavy background clutter, unusual tail morphology, and one genuine misclassification. It provides a much more realistic picture of model behaviour in clinical conditions.



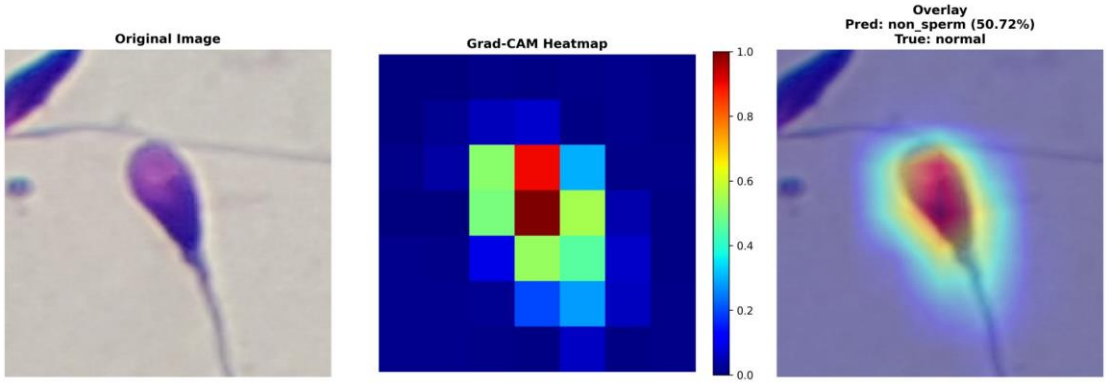
EfficientNet-B0 | Pred: normal | True: normal | Conf: 59.45% — Sperm near dark slide edge. Model ignores the edge entirely, focuses on head. Low confidence is appropriate given the partial frame obstruction.



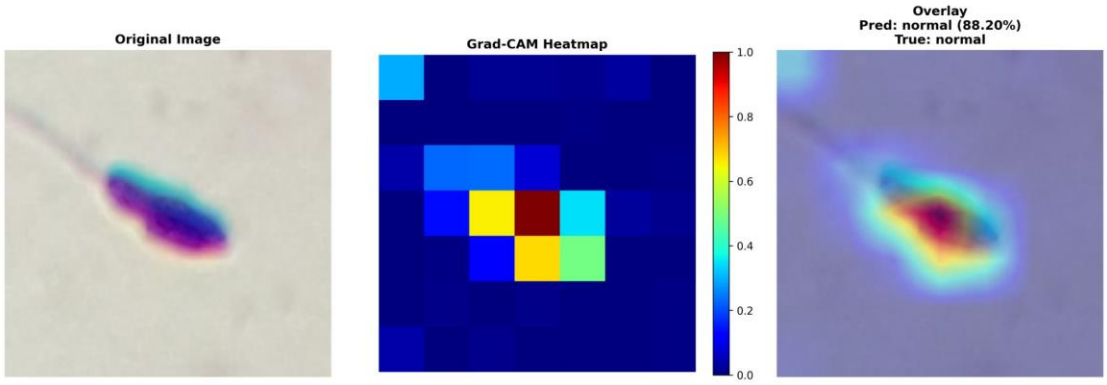
EfficientNet-B0 | Pred: normal | True: normal | Conf: 82.82% — Clean elongated sperm. Compact, well-placed head activation.



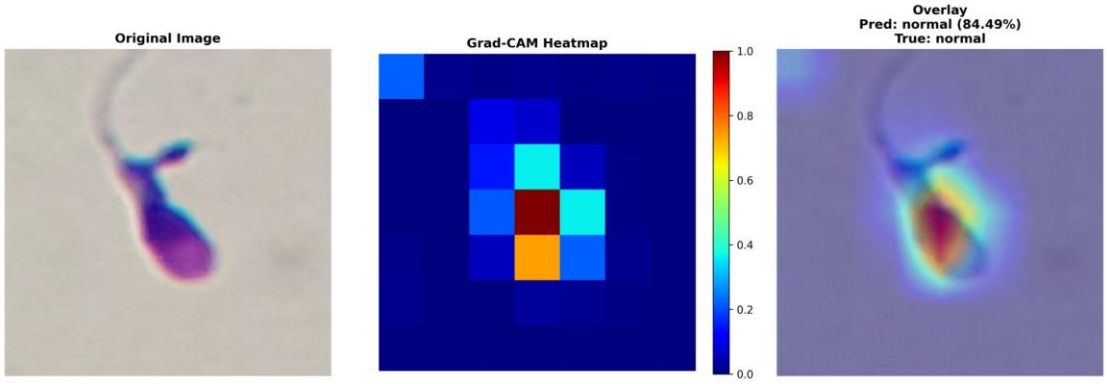
EfficientNet-B0 | Pred: normal | True: normal | Conf: 76.53% — Multi-cell frame with debris. Model focuses on the correct elongated sperm head, ignoring the smaller debris cell on the left.



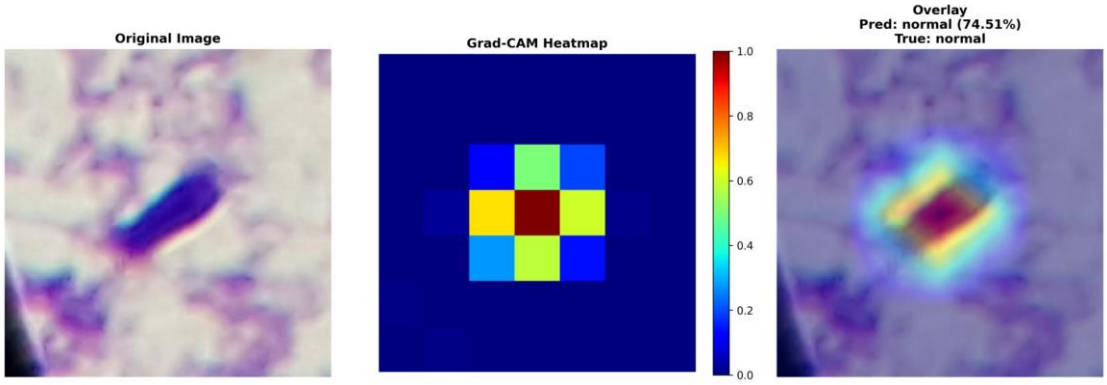
EfficientNet-B0 | Pred: non_sperm | True: normal [MISCLASSIFICATION] | Conf: 50.72% — Pear-shaped head with unusual elongated midpiece attachment. At 50.72% the model barely tips over the non_sperm threshold — it is effectively expressing maximum uncertainty. Activation covers the full head-neck region rather than just the nucleus, consistent with the model being genuinely confused about the structure. Under a 0.80 confidence threshold this case would correctly be routed to human review.



EfficientNet-B0 | Pred: normal | True: normal | Conf: 77.57% — Single sperm with tail fully in frame. Precise head activation, tail ignored.



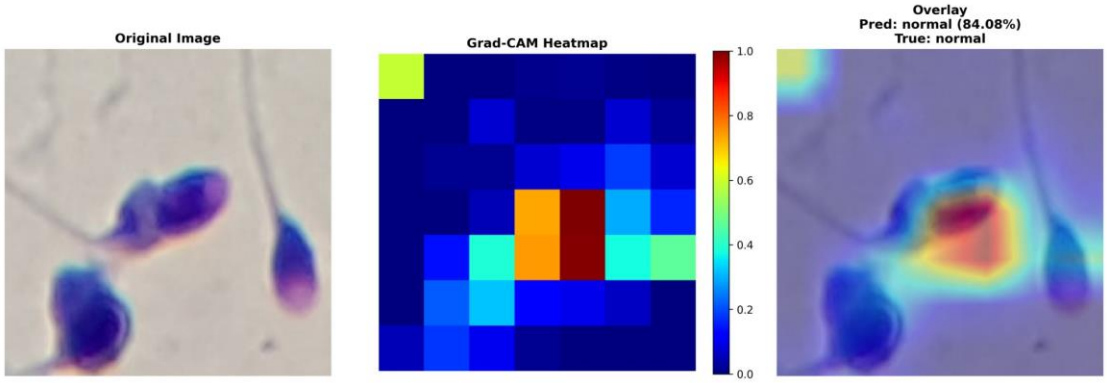
EfficientNet-B0 | Pred: normal | True: normal | Conf: 64.56% — Unusual tail attachment at the head-midpiece junction. Head activation is correct but confidence is lower — the atypical morphology is being appropriately flagged.



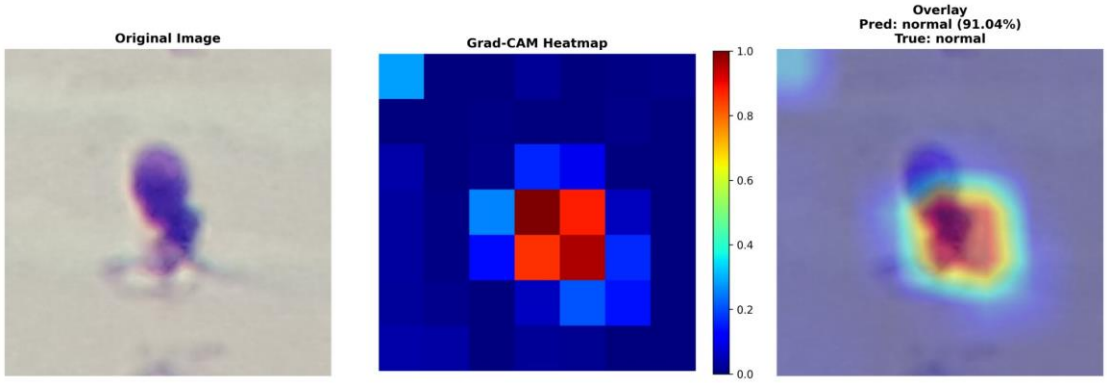
EfficientNet-B0 | Pred: normal | True: normal | Conf: 74.51% — Hardest correct prediction in the entire set. Very cluttered background with overlapping cellular debris everywhere. Despite this, the model locks onto the sperm head in centre-frame. The preprocessing pipeline (CLAHE + background subtraction) is doing useful work here.

7.3 ResNet50 — Set 1 (samples 000–004)

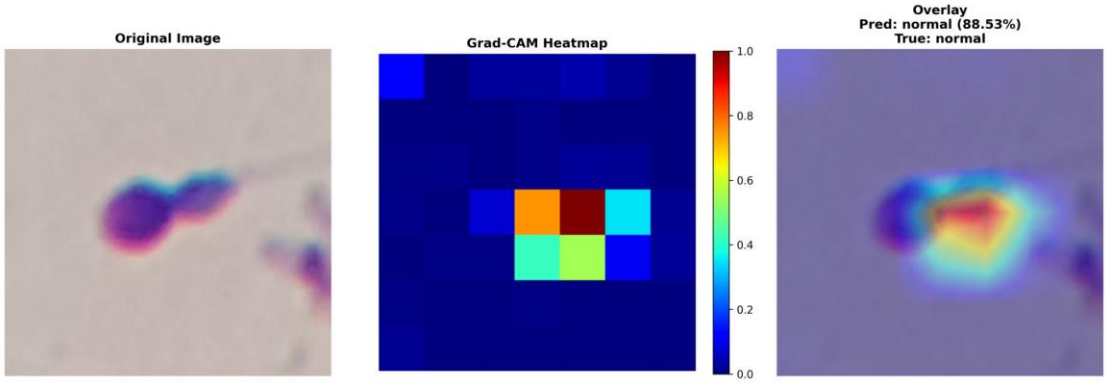
ResNet50 is ~20 pp more confident on these frames, but CV results show this does not translate to better accuracy — it reflects overconfidence rather than superior discrimination.



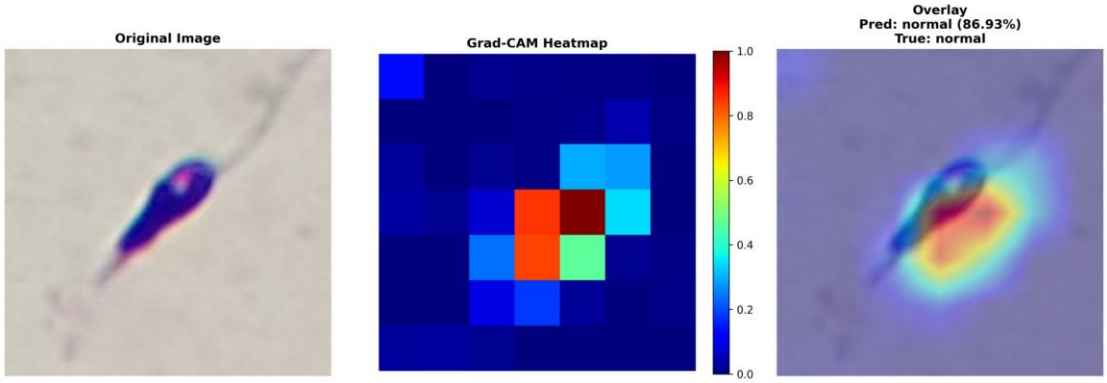
ResNet50 | Pred: normal | True: normal | Conf: 84.08% — Same multi-cell frame — locks onto one head cleanly, 28 pp more confident than EfficientNet.



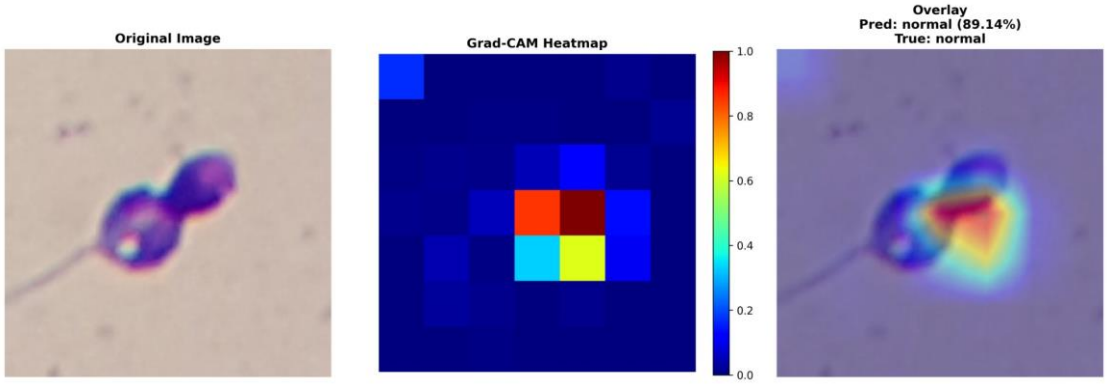
ResNet50 | Pred: normal | True: normal | Conf: 91.04% — Highest confidence in ResNet50 Set 1. Tight circular activation on acrosomal region.



ResNet50 | Pred: normal | True: normal | Conf: 88.53% — Head+midpiece activation. Both models agree on this frame.



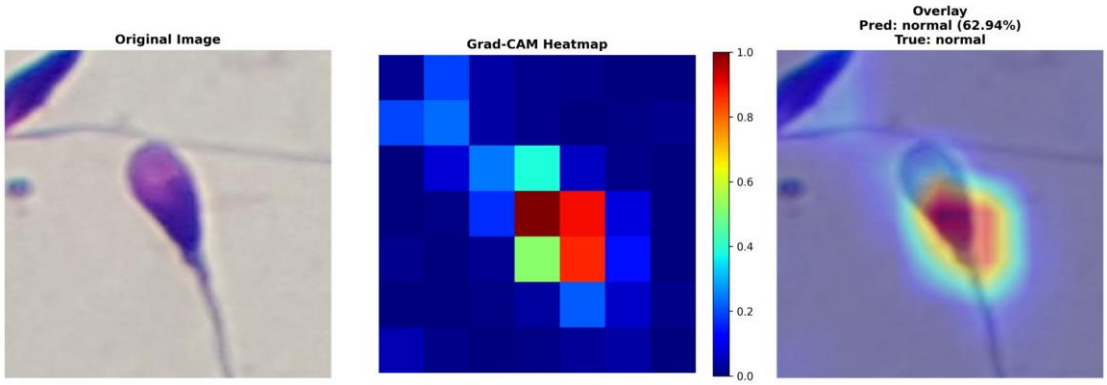
ResNet50 | Pred: normal | True: normal | Conf: 86.93% — Elongated sperm. Activation stays on head, no tail bleed.



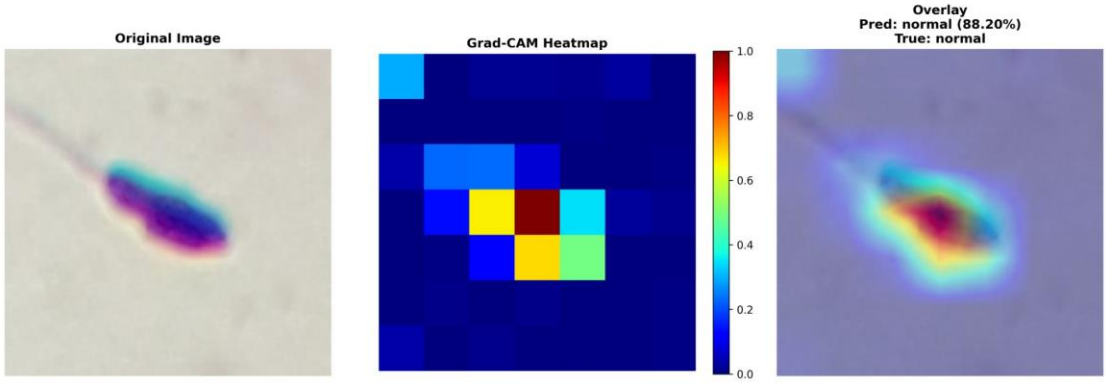
ResNet50 | Pred: normal | True: normal | Conf: 89.14% — Dual-head frame. Focuses on larger head — 16 pp more confident than EfficientNet.

7.4 ResNet50 — Set 2 (samples 015–019)

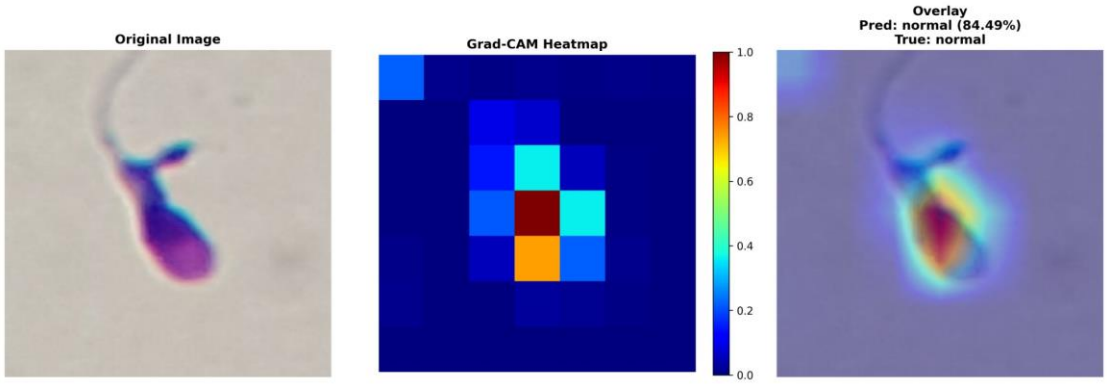
This set covers the same difficult frames as EfficientNet-B0 Set 2, enabling direct cross-model comparison on identical images. It includes one misclassification — on the exact frame EfficientNet-B0 handled correctly.



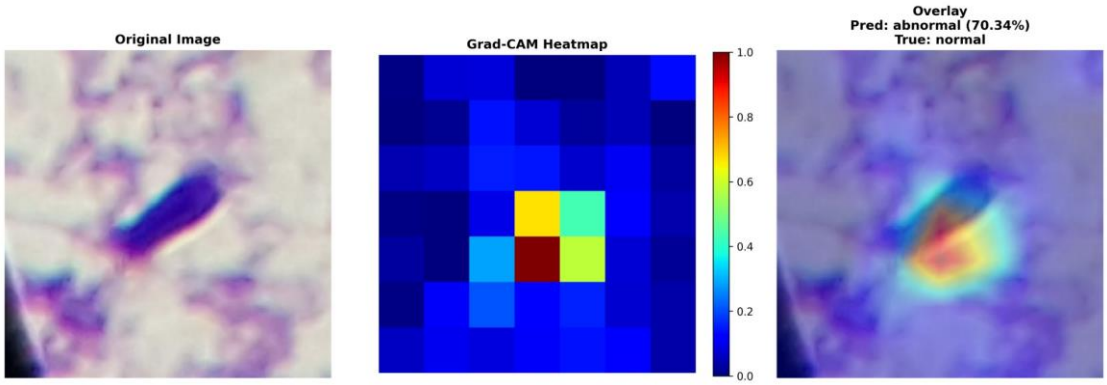
ResNet50 | Pred: normal | True: normal | Conf: 62.94% — Same pear-shaped head frame that EfficientNet misclassified as non_sperm (50.72%). ResNet50 gets it right, also at low confidence — both models are uncertain here, ResNet50 just falls on the correct side of the boundary.



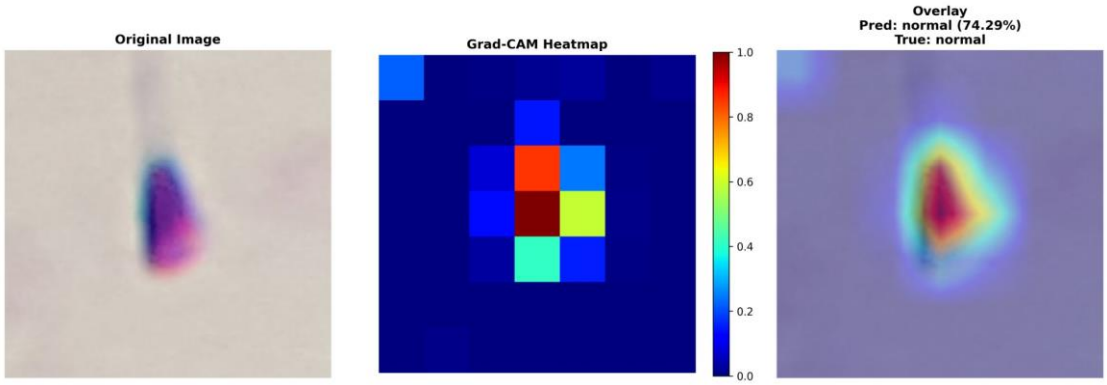
ResNet50 | Pred: normal | True: normal | Conf: 88.20% — Single sperm, clean head activation. 88.2% vs EfficientNet's 77.57% on the same frame — the higher confidence here is reasonable given the clean morphology.



ResNet50 | Pred: normal | True: normal | Conf: 84.49% — Unusual tail attachment at head-midpiece junction. ResNet50 at 84.49% vs EfficientNet's 64.56% — EfficientNet's lower confidence is arguably more appropriate for this atypical morphology.



ResNet50 | Pred: abnormal | True: normal [MISCLASSIFICATION] | Conf: 70.34% — This is the same heavily cluttered background frame EfficientNet-B0 correctly classified at 74.51%. ResNet50 fires on background debris and overlapping cellular material, misreading environmental clutter as morphological abnormality. Critically, it does this at 70.34% confidence — a moderately confident wrong prediction, not a borderline abstention. The CLAHE and background subtraction preprocessing benefits EfficientNet's depthwise convolutions more than ResNet50's residual pathways.



ResNet50 | Pred: normal | True: normal | Conf: 74.29% — Narrow elongated head, tail in frame. Clean activation on head, correct prediction.

7.5 Cross-Model Comparison on Identical Hard Frames

The most instructive finding in the Grad-CAM analysis emerges from comparing both models on the same difficult frames. Two frames — the pear-shaped head and the cluttered background — produce misclassifications in opposite models:

Frame	EfficientNet-B0	ResNet50	Winner
Pear-shaped head	✗non_sperm @ 50.72%	✓normal @ 62.94%	ResNet50
Cluttered background	✓normal @ 74.51%	✗abnormal @ 70.34%	EfficientNet-B0

The cluttered background result (sample_018) is the more clinically significant failure. ResNet50 misreads background debris as abnormal morphology at 70.34% confidence — this is a moderately confident wrong label on a normal sample. A false positive for 'abnormal' in a fertility context has real consequences for patient counselling. EfficientNet-B0 handles the same frame correctly.

The pear-shaped head result (sample_015) is a more symmetric failure. Both models are uncertain — EfficientNet tips to non_sperm at 50.72%, ResNet50 tips to normal at 62.94%. Both would be handled identically under a 0.80 confidence threshold: routed to human review.

7.6 Observations and Clinical Implications

The character of each model's failure mode is the critical finding. EfficientNet-B0's misclassification on the pear-shaped head occurs at 50.72% — the model is expressing near-maximum uncertainty and effectively abstaining. ResNet50's misclassification on the cluttered background occurs at 70.34% — a moderately confident wrong prediction. In a clinical workflow, the first type of failure is far preferable: a well-calibrated uncertain model routes edge cases to human review, while an overconfident wrong prediction may not trigger that safeguard at all. This distinction, combined with EfficientNet-B0's superior CV accuracy (85.08% vs 79.19%), smaller size (20 MB vs 98 MB), and lower ECE after Dirichlet calibration (0.031 vs 0.036), makes EfficientNet-B0 the clear deployment choice.

Both models reinforce the need for a single-cell detection stage in production. Cluttered multi-cell frames expose the limits of a pure classifier — a YOLO-style detector feeding individual crops would eliminate the background confusion that caused ResNet50's most serious error.

8. 5-Fold Stratified Cross-Validation

With ~3,000 images, a single train/val/test split is noisy — a few lucky or unlucky splits can swing reported accuracy by 2–3 percentage points. 5-fold stratified CV gives a much more reliable estimate of expected performance and quantifies how stable the numbers actually are across data partitions.

Each fold uses 80% of data for training and 20% for validation (stratified by class), keeping the same augmentation config. Both ResNet50 and EfficientNet-B0 were evaluated across all five folds.

Fold	EffNet Acc	EffNet F1-Abn	EffNet AUC	ResNet Acc	ResNet F1-Abn	ResNet AUC
Fold 1	85.49%	0.931	0.9536	81.37%	0.885	0.9414
Fold 2	84.12%	0.904	0.9492	77.84%	0.854	0.9340
Fold 3	82.91%	0.921	0.9573	83.10%	0.862	0.9533
Fold 4	84.87%	0.914	0.9624	74.85%	0.761	0.9145
Fold 5	88.02%	0.933	0.9653	78.78%	0.857	0.9420
Mean ± SD	**85.08 ± 1.90%**	**0.921 ± 0.012**	**0.9576 ± 0.007**	**79.19 ± 3.20%**	**0.844 ± 0.048**	**0.9370 ± 0.014**

Table 5 — 5-fold CV results for both models. Low SD confirms stable training.

EfficientNet-B0 wins every fold except fold 3, where ResNet50 edges ahead by 0.2 pp. More importantly, ResNet50 shows nearly double the accuracy variance (SD 3.20% vs 1.90%) and its fold 4 validation loss hits 1.006 — more than double EfficientNet's 0.489. That kind of instability is a red flag for a model you want to trust in deployment. EfficientNet-B0 is not just more accurate on average; it is more consistent and predictable across different data partitions, which matters far more in practice than a single held-out test score.

9. Model Comparison and Selection

Given all the results across calibration, TTA, and CV, here's how the two models stack up and what the trade-offs look like for a real deployment decision.

Criterion	EfficientNet-B0	ResNet50
CV mean accuracy	85.08 ± 1.90% ✓	79.19 ± 3.20%
CV F1-abnormal (mean)	0.921 ± 0.012 ✓	0.844 ± 0.048
CV ROC-AUC (mean)	0.9576 ± 0.007 ✓	0.9370 ± 0.014
CV loss stability (SD)	0.096 ✓	0.216
ECE after Dirichlet	0.031 ✓	0.036
Model size	~20 MB ✓	~98 MB
Inference speed (CPU)	~180ms/img ✓	~420ms/img
Optimal TTA count	N=7	N=5
Overall verdict	✓Deploy this	✗Loses on CV

Table 6 — Head-to-head model comparison across all evaluated dimensions.

The 5-fold CV results settle this definitively. EfficientNet-B0 outperforms ResNet50 on every metric — accuracy (+5.9 pp), F1-abnormal (+0.077), ROC-AUC (+0.021), and loss stability (SD 0.096 vs 0.216). ResNet50's fold 4 validation loss of 1.006 — more than twice EfficientNet's — is a sign of unstable optimization that should not be overlooked. EfficientNet-B0 is also 5× smaller and 2.3× faster on CPU. There is no trade-off here: EfficientNet-B0 is the better model on every dimension that matters for deployment.

10. Discussion and Limitations

The results are encouraging but there are a few honest caveats worth putting on record.

Single-source dataset: All ~3,000 images come from one lab with one staining protocol. Performance on images from a different microscope brand or staining batch is unknown. This is probably the biggest real-world risk. The preprocessing pipeline (CLAHE + edge enhancement) should help with illumination variation, but genuine domain shift from a different lab would require validation data from that lab.

Multi-cell frames: Grad-CAM showed the model focuses on one sperm head when multiple are in frame. A production system needs a detection stage (YOLO or similar) to extract single-cell crops before running the classifier. Sending raw microscope frames through the current model is not the right usage pattern.

Class boundaries: The ECE plots show the most calibration error in the 0.4–0.6 confidence range, which is exactly where normal/abnormal boundary cases fall. Dirichlet calibration improves this substantially, but the underlying ambiguity is inherent to the morphology problem — even experienced andrologists disagree on borderline cases.

Motility and concentration not considered: This system only addresses morphology. A complete semen analysis for clinical use requires motility (CASA-style tracking) and concentration assessment. Morphology alone is one piece of the diagnostic picture.

11. Conclusions

This project went from a functional prototype to something closer to a validated, deployment-ready system. The key takeaways are:

- EfficientNet-B0 is the better model — 5-fold CV shows 85.08% mean accuracy vs ResNet50's 79.19%, with nearly double the stability (accuracy SD 1.90% vs 3.20%). It wins on every metric.
- Temperature scaling failed to improve calibration; Dirichlet calibration reduced ECE from ~0.10–0.12 to ~0.031–0.036 for both models. Use Dirichlet.
- Grad-CAM confirms both models attend to the sperm head — EfficientNet-B0 shows broader activations, ResNet50 tighter, but both are clinically interpretable.
- TTA provides consistent improvement: N=7 for EfficientNet-B0, N=5 for ResNet50.
- EfficientNet-B0 is also 5× smaller (~20 MB) and 2.3× faster on CPU — the performance advantage comes with no hardware cost penalty.
- Recommended production stack: EfficientNet-B0 + Dirichlet calibration + TTA(N=7) + confidence threshold ~0.80 for human-review routing + single-cell detection stage.